Chapter III.9

Noah Parker

9.1.

9.3. Some examples of flatness and nonflatness.

(a) If $f : X \to Y$ is a finite surjective morphism of nonsingular varieties over an algebraically closed field k, then k is flat.

(b) Let X be a union of two planes meeting at a point, each of which maps isomorphically to a plane Y. Show that f is not flat. For example, let $Y = \operatorname{Spec} k[x, y]$ and $X = \operatorname{Spec}[x, y, z, w]/(z, w) \cap (x + z, y + w)$.

(c) Again let $Y = \operatorname{Spec} k[x, y]$, but take $X = \operatorname{Spec} k[x, y, z, w]/(z^2, zw, w^2, xz - yw)$. Show that $X_{\text{red}} \cong Y$ has no embedded points, but that f is not flat.

9.4. *Open Nature of Flatness.*

9.5.

9.7.

9.10